# PROGRAMME

## Second SSCI Annual Combustion Meeting

*Theme: "Future Combustion: Science, Innovation and Sustainability"*

**16 November 2026**

Level 2 Theatrette, CREATE Tower
1 CREATE Way, Singapore 138602

**Organized by:**

Singapore Section of the Combustion Institute (SSCI)

**Hosted by:**

Cambridge Centre for Advanced Research and Education in Singapore (CARES)

# 1 Welcome Message

The Singapore Section of the Combustion Institute (SSCI) is delighted to welcome you to the **Second SSCI Annual Combustion Meeting**, held on **16 November 2026** at the Level 2 Theatrette, CREATE Tower, Singapore. Following the highly successful inaugural meeting, the second annual meeting serves as a forum for researchers, practitioners, and students from academia, industry, and government to present recent advances in combustion research, exchange ideas, and establish new collaborations. Under the theme **"Future Combustion: Science, Innovation and Sustainability"**, the programme features invited and contributed oral presentations together with a poster session showcasing emerging research and innovations.

This year's meeting covers all areas of combustion science, including but not limited to:

- **Fundamental Combustion Science:** *chemical kinetics, flame dynamics, turbulent combustion, combustion instabilities, diagnostics*
- **Advanced Combustion Technologies:** *engines, gas turbines, industrial combustion, propulsion systems, novel combustion concepts*
- **Sustainable and Low-Carbon Combustion:** *hydrogen, ammonia, sustainable aviation fuels, biofuels, e-fuels, carbon-neutral combustion*
- **Combustion Modelling, Simulation, and Control:** *CFD, reacting flow modelling, combustion control, AI/data-driven methods, digital twins*
- **Emissions, Fire, and Energy Systems:** *pollutant formation and mitigation, fire safety, battery fires, carbon capture, integrated energy systems*

We hope that this meeting inspires meaningful scientific exchanges, strengthens collaborations, and continues to build a vibrant and impactful combustion research community in Singapore.

## Organizing Committee

### Organizers:

- **Prof. Huangwei Zhang** (NUS)
- **Dr. Yichen Zong** (CARES)
- **Dr. Wai Lee Chan** (SUTD)

### Secretaries:

- **Dr. Tong Su** (CARES)
- **Dr. Victor Lafaurie** (CARES)

# 2 Programme Overview

| Time | Activity |
|---|---|
| 08:30 – 08:35 | **Welcome Address** by **Prof. Huangwei Zhang** *(SSCI Chairman)* |
| 08:35 – 08:50 | **Opening Speech** by **Mr. John Chen** *(Guest of Honour, Advanced Energy and Decarbonisation Office)* |
| 08:50 | **Group Photo** |
| | **Session I: Emerging Research from Singapore** |
| 08:50 – 09:20 | KEYNOTE **Prof. Markus Kraft** *(University of Cambridge / CARES)*<br>**Molecules to Nanoparticles: Soot Inception Challenges in the Age of Alternative Fuels** |
| 09:20 – 09:40 | **Dr. George Xu** *(A*STAR IAIC)*<br>**Modeling & Simulation of Battery-Related Thermal Management and Fire Risks Using CFD** |
| 09:40 – 10:00 | **Dr. Xinyi Zhou** *(National University of Singapore)*<br>**Ammonia Engine Development for Maritime Decarbonization** |
| 10:00 – 10:20 | **Dr. Hongxun Xiao** *(National University of Singapore)*<br>**Picosecond Laser Electronic Excitation Tagging Velocimetry: Toward Combustion and Reacting-Flow Diagnostics** |
| 10:20 – 11:00 | **Coffee Break & Poster Session** |
| | **Session II: Hydrogen and Ammonia Energy** |
| 11:00 – 11:30 | KEYNOTE **Prof. Yiguang Ju** *(Princeton University)*<br>**Challenges and Opportunities of Ammonia Combustion for Power Generation** |
| 11:30 – 11:50 | **Prof. Dong Han** *(Shanghai Jiao Tong University)*<br>**Combustion Fundamentals and Engine Combustion Technologies for Zero-Carbon Fuels** |
| 11:50 – 12:10 | **Dr. Stone Zhang** *(Amogy)*<br>**Integrated Ammonia-to-Power System Enabled by Advanced Ammonia Cracking Technology** |
| 12:10 – 12:30 | **Mr. Hongdou Huang** *(Baker Hughes)*<br>**From Combustion Innovation to Commercial Deployment: Market Perspectives on Hydrogen and Ammonia Gas Turbines** |
| 12:30 – 14:00 | **SSCI AGM, 12:30 – 13:10** ***(SSCI members only)***<br>**Lunch & Poster Session** |

| Time | Activity |
|---|---|
| | **Session III: Explosion, Detonation and AI** |
| 14:00 – 14:30 | KEYNOTE **Dr. Nabiha Chaumeix** *(CNRS-Ingénierie, ICARE)*<br>**H2 Explosion Risk Assessment: State of Knowledge and Database from Nuclear Community (online)** |
| 14:30 – 14:50 | **Prof. Zheng Chen** *(Peking University)*<br>**Detonation Propagation in Non-Uniform Mixtures** |
| 14:50 – 15:10 | **Dr. Yunpeng Yang** *(Shanghai Fire Science and Technology Research Institute of MEM)*<br>**Multi-Phased Combustion and Explosion Hazard Development of Hydrogen Releases: Coupling Effects of Wind Condition, Ignition Sequence and Obstacle Perturbation** |
| 15:10 – 15:30 | **Dr. Chengyu Wu** *(Rankyee Technology)*<br>**Towards Physical AI for Multi-Regime Combustion: High-Fidelity Simulation Datasets, Neural FGM and Applications** |
| 15:30 – 15:40 | **Young Researcher Award Ceremony** |
| 15:40 – 16:10 | **Coffee Break & Poster Session** |
| | **Session IV: Engine Combustion and Fuels** |
| 16:10 – 16:30 | **Prof. Youhi Morii** *(Tohoku University)*<br>**Recent Progress in Predicting Engine Knock from the Ignition-Flame Relationship** |
| 16:30 – 16:50 | **Prof. Gang Wu** *(Shanghai Maritime University)*<br>**Recent Progress in the Promotion of Marine Alternative Fuel and Engine Technologies** |
| 16:50 – 17:10 | **Dr. Muhamad Syiham Muez Ramli** *(School of Mechanical Engineering, Universiti Sains Malaysia)*<br>**Image Processing-Based Evaluation of Flame Geometry and Brightness in Diluted LPG Combustion** |
| 17:10 – 17:30 | **Prof. Appathurai Venugopalan Sowriraajan** *(Fire and Combustion Research Center, JAIN University, India)*<br>**Combustion of Polymeric and Biomass Wastes** |
| 17:30 | **Oral Presentation and Poster Award Ceremony**<br>**Closing Remarks** |

# 3 Guest of Honour & Invited Speakers

## Guest of Honour

### Mr. John Chen

*Advanced Energy and Decarbonisation Office (AEDO)*



**Biography:** John Chen leads AEDO's R&D team, which assesses R&D proposals in the TRL 1–6 range for funding and actively manages funded projects to ensure they deliver technical progress with clear pathways to scale and commercialisation. He also oversees partnership-building efforts across industry, from local facility owners and equipment manufacturers to startups and venture capitalists.

John concurrently serves on the Boards of the Centre for Hydrogen and Carbon Innovations (CHCI), Energy Research Institute at NTU (ERI@N), and Solar Energy Research Institute of Singapore (SERIS).

Prior to AEDO, John spent over a decade in policy roles at METI, most recently at the intersection of climate and economic policy. He holds a Chemical Engineering degree from the National University of Singapore and an MBA from Nanyang Technological University.

Keynote Lecture

# H2 Explosion Risk Assessment: State of Knowledge and Database from Nuclear Community

**Dr. Nabiha Chaumeix**

*CNRS-Ingénierie, ICARE*

**Abstract:** A key priority in global hydrogen deployment strategies is the development and scaling of carbon-free hydrogen production pathways. Among the leading technologies being pursued, water electrolysis stands out as a cornerstone for achieving deep decarbonization of hydrogen production by 2050. Several electrolysis technologies exist, but alkaline and proton exchange membrane systems are currently the most mature, representing roughly 60% and 22% of the global market in 2023, respectively. Despite their promise, electrolysis systems are subject to specific safety challenges related to hydrogen leaks. The nuclear community has developed, over decades of research, a substantial knowledge spanning for ignition of slow flames to the generation of fast flames eventually leading to detonation, but also on mitigation methods, passive and active measures.

The present talk will focus on describing the key feature of hydrogen-explosion safety with an overview of the different combustion regimes and map the explosion hazard from subsonic to supersonic combustion and the impact on structures. Examples based on ICARE's work on hydrogen safety assessment and mitigation methods will be given.



**Biography:** Dr. Nabiha CHAUMEIX, director of ICARE, a full-body CNRS laboratory, has a Ph.D. in mechanical engineering (University of Orléans, 1993). She was president of the Institute of Dynamics of Explosions and Reactive Systems (IDERS, 2017-2022) and deputy director of the French Research Network on Soot. Dr Chaumeix has over 25 years of experience in the flame dynamics, explosions safety and combustion chemical kinetics. The research developed by Dr Chaumeix is related to: (i) high temperature chemical kinetics using shock tubes and has devoted more than a decade in the study of soot formation from heavy fuels; (ii) the determination of the combustion fundamental properties such as flammability limits, laminar flame velocities, auto-ignition delay times, detonation characteristics (cell size, detonation speed, etc.) and the development of detailed chemical kinetics applicable to these phenomena; (iii) assessment of Safety explosion criteria with the detailed study of flame acceleration covering both subsonic and supersonic flames. The research is developed in the framework of National projects (ANR- HYDROMEL, MITHYGENE, IRSIS, SYTCOM, PHYSSA, PEPR-H2-ESKHYMO & AIDHY), European projects (SiA-TEAM, ARCHER, AMHYCO, FUN-PM, SASPAM-SA), International projects (EIG CONCERT-JAPAN 2021-STACY) and in several projects with different industries and institutions (TotalEnergies, IRSN, EDF, AREVA, Air Liquide, CNES, CEA, . . . etc.). She is a member of IEA TCP Task 43 "Safety and RCS of Large Scale Hydrogen Energy Applications" and of Task 44 "Hydrogen from Nuclear Energy".

Keynote Lecture

# Challenges and Opportunities of Ammonia Combustion for Power Generation

**Prof. Yiguang Ju**

*Princeton University*

**Abstract:** Ammonia is a potential hydrogen carrier for power generation because of its higher energy density and easier storage and transportation. However, ammonia slips, high NOx/N2O emissions, and fast metal surface nitridation have made it challenging in direct applications to existing gas turbine and diesel engine applications.

In this lecture, we will first discuss the potential impact of ammonia combustion on emissions, climate, and metal surface nitridation. Then, we will present the results of NOx/N2O of high pressure (100 atm) low temperature oxidation. The three important low temperature ammonia oxidation kinetic pathways and its impact on warm flames and detonation transition will be presented.

After that, direct measurements of critical radicals such as NHx/HO2 in ammonia combustion using Faraday rotational spectroscopy and femtosecond laser absorption will be discussed. The data will be used to determine the rates of important elementary reactions such as HO2+NH2 and examine the metal surface nitridation mechanism.

Finally, a new concept of high pressure fully premixed turbulent jet stirred ammonia combustion will be presented. The potential of plasma assisted ammonia combustion for low NOx emissions will also be discussed.



**Biography:** Yiguang Ju is the Robert Porter Patterson Professor in Mechanical and Aerospace Engineering, Associated Faculty of Princeton Plasma Physics Laboratory, and is the Director of DOE Energy Earthshot Research Center (EERC) for plasma assisted hydrogen production.

Ju's research interests include combustion, green fuels, plasma, and electrified manufacturing. Ju is a Fellow of The American Institute of Aeronautics and Astronautics (AIAA), The American Society of Mechanical Engineers (ASME), and The Combustion Institute.

He has received many awards such as the Friedrich Wilhelm Bessel Research Award from the Alexander von Humboldt Foundation, the AIAA propellant and combustion award, the Alfred C. Egerton Gold Medal from the Combustion Institute, the 2025 and 2026 R&D 100 Awards by R&D World magazine, and the international award from Tohoku University (2026).

Keynote Lecture

# Molecules to Nanoparticles: Soot Inception Challenges in the Age of Alternative Fuels

**Prof. Markus Kraft**

*University of Cambridge / CARES*

**Abstract:** Understanding soot inception, the self-assembly of gas-phase molecules into carbonaceous nanoparticles, remains a fundamental challenge in combustion science. Advances in mass spectrometry and atomic force microscopy reveal that simple physical dimerization of small aromatics is thermodynamically unviable at flame temperatures, rendering traditional kinetic models non-predictive. Instead, molecular growth proceeds via curved PAHs and large aromatic oligomers ("aromers") through physically stabilised soot inception, where localized radicals and reactive edges bridge physical and chemical routes.

Critically, the transition to alternative fuels does not eliminate this soot challenge. Co-firing hydrocarbons with low-carbon vectors like ammonia introduces complex chemical cross-talk, where partially cracked ammonia creates a non-linear competition between ammonia-driven soot inhibition and hydrogen-driven soot promotion alongside a distinct soot trade-off. Capturing these competing pathways within unified, physically stabilised inception frameworks represents a vital frontier for predictive modelling in future net-zero combustors.



**Biography:** Professor Kraft FREng leads the Computational Modelling Group at the Department of Chemical Engineering and Biotechnology. His research spans the computational modelling and optimisation of particle formation processes, with particular expertise in the combustion synthesis of organic and inorganic nanoparticles. Drawing on stochastic and deterministic methods, the group develops detailed mechanistic models that capture the complex physical and chemical processes governing particle nucleation, growth, and aggregation. This work has yielded significant advances in the understanding of soot formation, titania synthesis, and silicon nanoparticle production, with applications across the automotive, power, and chemical industries.

He co-edited the book "Intelligent Decarbonisation", which comprehensively assesses the current and future impact of digital technologies and artificial intelligence on the decarbonisation of key economic sectors. He is a Fellow of Churchill College and a Professor in the Department of Chemical Engineering and Biotechnology at the University of Cambridge. Until the beginning of 2025 he was the founding Director of the Cambridge Centre for Advanced Research and Education in Singapore (CARES), the University of Cambridge's first overseas research centre, based at CREATE. In 2025 Prof. Kraft is a visiting Professor in the Department of Chemical Engineering at MIT. He is an elected Fellow of the Royal Academy of Engineering.

Invited Talk

# Detonation Propagation in Non-Uniform Mixtures

**Prof. Zheng Chen**

*Peking University*

**Abstract:** Understanding detonation propagation in non-uniform mixtures is essential for rotating detonation engines, hypersonic propulsion, and explosion safety. This talk explores the dynamics and critical conditions governing detonation propagation in spatially inhomogeneous reactive mixtures. Recent advances are presented, with a focus on thermal stratification, inert layers, and multi-dimensional mixture non-uniformity. The effects of non-uniformity on detonation initiation, quenching, reinitiation, propagation speed, and cellular structure are examined.

In hypersonic reactive flows, thermal stratification can produce autoignition-driven reaction fronts, oblique detonation waves, or decoupled shock/reaction fronts, depending on the temperature gradient. For detonation across inert layers, successful reinitiation occurs only below a critical layer thickness; and transverse shock interactions induce local autoignition/explosion and overdriven detonation. Besides, a double cellular structure is observed for relatively large spacing between inert layers. The formation of double cellular structure is interpreted.

**Biography:** Dr. Zheng Chen is currently a Bo-Ya Distinguished Professor at Peking University. He received his bachelor's and master's degrees from Tsinghua University and his Ph.D. from Princeton University. His research focuses on flame dynamics in fundamental combustion processes. By combining theoretical analysis, numerical simulation, and experimental measurement, he has worked on challenging problems in ignition theory, accurate laminar flame speed measurement, deflagration-to-detonation transition, and detonation engines.



He was awarded the Hiroshi Tsuji Early Career Researcher Award from the Combustion Institute. He received the Distinguished Paper Award at the 39th International Symposium on Combustion and delivered a plenary lecture at the 41st International Symposium on Combustion. He is a Fellow of the Combustion Institute and currently serves on its Board of Directors. He is an associate editor of Combustion and Flame.

Invited Talk

# Combustion Fundamentals and Engine Combustion Technologies for Zero-Carbon Fuels

**Prof. Dong Han**

*Shanghai Jiao Tong University*

**Abstract:** This speech focuses on the combustion fundamentals and combustion technologies of zero-carbon ammonia fuel in compression-ignition engines. Combustion reaction kinetics model for binary fuels with high ammonia proportions was developed, enabling high-accuracy prediction of the fundamental combustion characteristics of ammonia fuel and engine combustion processes. Combustion control methods for ammonia-fueled engines based on the coordinated control of fuel reactivity and ignition strategy was proposed, providing a technical guidance for the efficient and stable operation of ammonia-fueled engines.

**Biography:** Dong Han is professor at Shanghai Jiao Tong University, and he also serves as vice dean of College of Smart Energy there. He is engaged in engine combustion and emissions control technology, and combustion theory and technology of low-carbon/zero-carbon fuels.



He has published 160 peer-reviewed journal papers and been granted 12 invention patents in China. He received the First Prize for Natural Science from the Chinese Society for Internal Combustion Engines. He serves as editorial board members of international journals in combustion and energy field, such as Applied Thermal Engineering and Engineering Energy.

Invited Talk

# From Combustion Innovation to Commercial Deployment: Market Perspectives on Hydrogen and Ammonia Gas Turbines

**Mr. Hongdou Huang**

*Baker Hughes*

**Abstract:** As global industries and power systems pursue net-zero ambitions, hydrogen and ammonia are emerging as important fuels for the future of power generation. Significant progress has been made in combustion technologies, enabling higher hydrogen blending ratios and reducing overall emissions. However, the pace of market adoption will ultimately depend on economics, infrastructure readiness, policy support, and evolving customer requirements.

This presentation will provide a brief industry perspective on the commercialization of hydrogen and ammonia gas turbines, drawing on market developments across power generation, industrial energy applications, and emerging hydrogen value chains. It will explore priority market segments, key adoption drivers, customer expectations, deployment challenges, and the role of gas turbine technology in enabling a practical and scalable energy transition. The presentation will also highlight lessons learned from early demonstration and commercial projects, and provide an outlook on how industry stakeholders can bridge the gap between technology readiness and large-scale commercial implementation.



**Biography:** Hongdou Huang is Senior Advisor for Hydrogen Market Development & Solution Architecture at Baker Hughes, where he leads strategic partnerships, commercial initiatives, and market development efforts to advance hydrogen solutions across the Asia-Pacific region.

With over 17 years of experience in the energy, power generation, and infrastructure sectors, Hongdou has built a career at the intersection of technology, commercialization, and business strategy. Since joining Baker Hughes in 2019, he has supported the growth of NovaLT gas turbine power solutions before moving into his current hydrogen-focused role supporting the energy transition. Prior to Baker Hughes, Hongdou held strategy and business development positions with leading OEM, EPC, IPP, and infrastructure organizations, including PW Power Systems (MHI), Abengoa, and Keppel Infrastructure. His experience working with government agencies, utilities, and financial institutions across Asia-Pacific provides him with a practical commercial perspective on hydrogen deployment and decarbonization.

He holds a degree in Environmental Science and Engineering from the National University of Singapore.

Invited Talk

# Recent Progress in Predicting Engine Knock from the Ignition-Flame Relationship

**Prof. Youhi Morii**

*Tohoku University*

**Abstract:** Knock limits the compression ratio, and hence the thermal efficiency, of spark-ignition engines. Conventional measures such as the octane number are relative and empirical, while integral methods require assumptions of homogeneity, single-stage ignition and history independence.

Our theory of the explosive transition of deflagration approaches knock from a different direction, by asking where the limit of flame existence lies and under what condition the reaction turns explosive, conditions governed mainly by temperature. Building on this theory, a hierarchy of systems, from a practical engine through a constant-volume chamber to one-dimensional direct numerical simulation, is connected by the pressure-temperature history of the end gas, reducing the engine to lower-dimensional systems from first principles so that the theoretical picture can be tested directly against engine conditions.

Recent progress and the outlook for predicting knock onset are discussed.



**Biography:** Youhi Morii is an Associate Professor at the Institute of Fluid Science, Tohoku University. He received his Ph.D. from the Graduate University for Advanced Studies (SOKENDAI) in 2012, and worked at the Japan Aerospace Exploration Agency and Waseda University before joining Tohoku University in 2018.

His research combines large-scale simulation with reaction-wave theory to describe autoignitive waves spanning deflagration, spontaneous propagation and detonation, with applications to engine knock and supersonic combustion. He received the Young Researcher Award of The Combustion Society of Japan in 2023 for theoretical and numerical work on the explosive transition of deflagration, and the JAXA President's and Director's Awards in 2014 and 2016.

Invited Talk

# Towards Physical AI for Multi-Regime Combustion: High-Fidelity Simulation Datasets, Neural FGM and Applications

**Dr. Chengyu Wu**

*Rankyee Technology*

**Abstract:** High-fidelity simulation is a key enabler of Physical AI for multi-regime combustion, yet industrial-scale dataset generation remains constrained by computational cost.

Rankyee addresses this challenge with DIMAXER, which integrates high-order Space-Time Expansion of Kinetic Energy Preserving Flux Reconstruction (STE-KEP-FR) with GPU computing and compressible flamelet-generated manifold (FGM) modeling. An oscillation-free formulation suppresses spurious numerical oscillations in compressible variable-density flows, while multi-dimensional FGM incorporates heat loss and multi-stream mixing. Neural FGM replaces high-dimensional table lookup with neural approximations of flamelet-based thermochemical mappings, reducing storage requirements as manifold dimensionality increases.

Case studies include Full-Annulus Combustor, Coupled Full-Engine Simulation, and Reusable Heavy-Lift Launch Vehicle Simulation. The full-annulus FGM large-eddy simulation (LES) uses 2 billion solution points on 15 GPUs, while the reusable launch vehicle simulation reaches 10 billion solution points. KJ66 Combustion Simulation further demonstrates a 400-million-point LES on eight consumer-grade GPUs.

These results demonstrate the computational feasibility of engineering-scale data generation. Rankyee outlines a path toward scalable production of spatially and temporally resolved flow and thermochemical data through workflows assisted by large language models and cloud computing, supporting combustion AI and digital twins.

**Biography:** Chengyu Wu is a CFD Application Engineer at Rankyee Technology, specializing in combustion simulation and GPU-accelerated high-fidelity modeling.



Rankyee Technology develops GPU-accelerated high-fidelity CFD software for Physical AI. Its flagship product, DIMAXER, is built on the proprietary Space-Time Expansion Kinetic Energy Preserving Flux Reconstruction (STE-KEP-FR) framework, enabling high-order, energy-stable simulations of unsteady compressible flows on consumer-grade GPUs.

In the combustion domain, Rankyee has developed a compressible FGM (Flamelet Generated Manifold) suite featuring oscillation-free numerics and multi-dimensional tabulation, further accelerated by neural network surrogates (Neural FGM).

Building on these capabilities, Rankyee has demonstrated billion-point, time-accurate combustion LES across a broad range of flow regimes — including full-annular gas turbine combustors, reusable launch vehicle nozzle exhaust, coupled full-engine configurations, and large-scale industrial furnaces — at a fraction of conventional supercomputing costs, providing scalable full-field data for Physical AI training and industrial design optimization.

Invited Talk

# Recent Progress in the Promotion of Marine Alternative Fuel and Engine Technologies

**Prof. Gang Wu**

*Shanghai Maritime University*

**Abstract:** Driven by the urgent need for maritime decarbonization, the exploration of zero- and low-carbon marine alternative fuels has become a focal point of global research. This lecture presents the recent advancements in marine alternative fuel and engine technologies, focusing on the fundamental combustion science and applied engineering innovations.

The talk will detail the transition pathways toward fuels such as biofuels, methanol, and ammonia in the maritime sector. Special attention will be given to the latest breakthroughs in the development of large ship engines, including advanced injection strategies, new design concepts for nozzle holes, the application of dedicated lubricant and fuel additives, as well as in-cylinder emission control technologies.

Ultimately, the lecture will summarize the current technological readiness and future research directions for developing highly efficient and clean marine power systems.



**Biography:** Prof. Gang Wu is an Associate Professor at Shanghai Maritime University, where he leads key initiatives in sustainable shipping. He serves as the Director of the Green Power Research Division at the International Research Center for Future Marine Technology and heads the Ship Power Department at the Merchant Marine College.

Dedicated to maritime decarbonization, his research centers on alternative marine fuel combustion and the operation and maintenance of low-speed marine engines. By bridging fundamental combustion science with applied engineering, Dr. Wu's work provides deep insights into the viability and application potential of biofuels, methanol, and ammonia for the future of green shipping.

Invited Talk

# Modeling & Simulation of Battery-Related Thermal Management and Fire Risks Using CFD

**Dr. George Xu**

*A*STAR IAIC*

**Abstract:** Effective thermal management and mitigation of fire risks associated with unexpected thermal runaway are critical to the safe and reliable deployment of lithium-ion batteries. This talk presents our recent research on computational fluid dynamics (CFD)-based modeling and simulation (M&S) of battery thermal management, thermal runaway, and fire dynamics, with particular emphasis on validating the underlying methodologies. Case studies include the thermal performance of a 48-cell battery pack, thermal runaway behavior in an 8-cell nickel manganese cobalt (NMC) battery pack, and an electric vehicle fire in a tunnel. These studies demonstrate how validated M&S tools can provide insights into heat transfer, thermal runaway propagation, and fire development. By enabling systematic evaluation of battery configurations, cooling strategies, and fire scenarios, these tools can support battery performance improvements, guide thermal management design, and assess the fire safety conditions arising from battery incidents, informing risk mitigation and emergency response planning.



**Biography:** Dr. George Xu (Xu Xiangguo) is a Senior Principal Scientist and Group Lead in the Sustainability Directorate at A*STAR's Institute of Advanced Intelligence and Computing (IAIC), Singapore. He holds a PhD in Mechanical Engineering from the National University of Singapore and brings around three decades of experience in experimental studies, computational fluid dynamics (CFD), and engineering applications. His work combines the development of computational methods with practical applications in heat transfer, ventilation, and fire and smoke propagation. During six years in engineering consultancy, he worked closely with multidisciplinary teams on thermal performance and fire safety assessments for buildings, rail infrastructure, and tunnels. His research interests centre on translating scientific advances into practical solutions that enhance the liveability and sustainability of Singapore's urban and built environments.

Invited Talk

# Integrated Ammonia-to-Power System Enabled by Advanced Ammonia Cracking Technology

**Dr. Stone Zhang**

*Amogy*

**Abstract:** Ammonia is emerging as a carbon-free energy carrier that is easier to store and transport than hydrogen, but turning it into reliable power at scale requires efficient, compact conversion technology. This presentation introduces Amogy, a company developing ammonia-to-power systems that decarbonize stationary, maritime, and heavy industries.

The presentation will describe Amogy's proprietary catalyst technology for ammonia conversion and show how it is integrated into a complete ammonia-to-power system, from ammonia feed through conversion to electricity generation. Will then present several application examples showing how the approach can be deployed across different power and industrial use cases. The talk concludes with perspectives on how ammonia-based power can support the transition to low-carbon energy systems.



**Biography:** Dr. Stone Zhang is the VP of Business Development at Amogy, where he leads business development and commercial operations in APAC. Stone had 10+ years of regional and global executive experience at DNV across the energy and maritime value chains. Right before Amogy, he also spent 5+ years in cross-border venture capital investing in deep-tech sectors between the Americas and Asia. Stone holds a PhD in Marine and Offshore Engineering from Harbin Engineering University (China) and an Executive MBA from Rice University (USA).

Invited Talk

# Ammonia Engine Development for Maritime Decarbonization

**Dr. Xinyi Zhou**

*National University of Singapore*

**Abstract:** Using carbon-free ammonia as a fuel is considered as a potential pathway for maritime decarbonization. Beyond concerns over future supply of green ammonia, whether ammonia engines themselves can deliver high efficiency and low emissions remains a central question. This presentation summarizes the recent development of marine ammonia engines, including the performance and emissions of different engine categories and combustion modes. It then shares technologies applicable to four-stroke medium-/high-speed engines and two-stroke low-speed engines, respectively. Finally, it discusses the potential of combustion similarity theory and scaled-model experiments to support the intensive development of ammonia engines.



**Biography:** Dr. Xinyi Zhou is currently a Senior Research Fellow at the National University of Singapore (NUS). He received his PhD from Shanghai Jiao Tong University (SJTU) in 2022. His research focuses on marine engine combustion and scaled-model experiments, with a recent emphasis on ammonia fuel marine engines. He serves as PI or Chief Scientist for several research projects in China and Singapore. His works on ammonia engine have been published as the first author in journals such as Joule and Nature Communications. He has received the SJTU Academic Star award and been selected for the Shanghai Rising-Star Program. He has been included in the world's top 2% scientists list and serves as a Young Editorial Board Member of Engineering.

# 4 Posters

## AI for Combustion

### Do Transformers Help? Machine Learning Screening of Biosynthetic Fuel Molecules

Yichen Zong[1,2], Yong Ren Tan[1]

[1] *Cambridge Centre for Advanced Research and Education in Singapore (CARES), Singapore*

[2] *3Y Energy Pte Ltd, Singapore*

### Posterior-Guided Adaptive Surrogate Modeling for Uncertainty Quantification and Optimization of Chemical Mechanisms Using Normalizing Flows

Hao Hu[1,2], Peng Ma[2], Shumeng Xie[1], Yicun Wang[1], Huangwei Zhang[1,2]

[1] *Department of Mechanical Engineering, National University of Singapore, Singapore*

[2] *Centre for Hydrogen and Carbon Innovations (CHCI), National University of Singapore, Singapore*

### A Physics-Constrained Parametric Reduced-Order Model with an Embedded Flamelet Manifold for Forced Ignition in Hydrogen/Air Counterflows

Yicun Wang[1], Shumeng Xie[1], Hao Hu[1], Zifeng Weng[1], Salim M. Shaik[2], Huangwei Zhang[1]

[1] *Department of Mechanical Engineering, National University of Singapore, Singapore*

[2] *Institute of Sustainability for Chemicals, Energy and Environment (ISCE[2]), A*STAR, Singapore*

### Dynamic Load-Balancing for GPU-Accelerated Chemistry Integration in OpenFOAM

Zifeng Weng[1,2], Yicun Wang[1], Huangwei Zhang[1]

[1] *Department of Mechanical Engineering, National University of Singapore, Singapore*

[2] *CNRS@CREATE, Singapore*

### From Molecular Kinetics to CFD: Supercritical Cracking of a Multicomponent Jet-A1 Surrogate for Regenerative Cooling

Jie-Yao Lyu[1,2], Changpeng Cai[2], Wenming Yang[1,2]

[1] *Centre for Hydrogen Innovations, National University of Singapore, Singapore*

[2] *Department of Mechanical Engineering, National University of Singapore, Singapore*

## Engine and Energy System

### Effect of Orifice Diameter on Jet Development for Turbulent Jet Ignition (TJI) Ammonia Engine

Zongkuan Liu[1,2], Haiqiao Wei[2], Wenming Yang[1]

[1] *Department of Mechanical Engineering, National University of Singapore, Singapore*

[2] *State Key Laboratory of Engines, Tianjin University, China*

### Ammonia as a Marine Fuel: Trapping $NO_x$ Emissions inside the Engine through Selective Non-Catalytic Reduction (SNCR)

Yingying Gao, Xinyi Zhou, Weixi Ni, Siyuan Dan, Ning Wang, Wenming Yang

*Department of Mechanical Engineering and Centre for Hydrogen Innovations, National University of Singapore, Singapore*

### Laminar Burning Characteristics and Emission Profiles of Premixed Ammonia/Hydrogen Mixtures under Engine-Relevant Conditions

Ning Wang[1], Xinyi Zhou[1], Yingying Gao[1], Siyuan Dan[1], Weixi Ni[1], Zongkuan Liu[1], Wenming Yang[1,2]

[1] *Centre for Hydrogen and Carbon Innovations (CHCI), National University of Singapore, Singapore*

[2] *Department of Mechanical Engineering, National University of Singapore, Singapore*

### Methane-Tar Synergistic Conversion during Partial Oxidation Process of High Tar-Concentration Syngas

Gao Ming[1], Cheng Liang[2], Wang Fuchen[2], Yang Wenming[1]

[1] *Centre for Hydrogen and Carbon Innovations (CHCI), National University of Singapore, Singapore*

[2] *State Key Laboratory of Coal Liquefaction, Gasification and Utilization with High Efficiency and Low Carbon Technology, East China University of Science and Technology, China*

### System Sizing and Dynamic Control of an Integrated Hydrogen Production Unit–CCGT System

Li Chin Law[1,3], Epaminondas Mastorakos[2]

[1] *Cambridge Centre for Advanced Research and Education in Singapore (CARES), Singapore*

[2] *Department of Engineering, University of Cambridge, United Kingdom*

[3] *EMICAST Pte. Ltd., Singapore*

## Ammonia and Hydrogen Flames

### Flame Structure and Lean Blow-Off Behaviour of Non-Premixed $NH_3/H_2/N_2$ Flames in a Laboratory-Scale Rich-Quench-Lean (RQL) Combustor

Tong Su[1], Jinhoon Choe[1], Epaminondas Mastorakos[1,2]

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[2] *Department of Engineering, University of Cambridge, United Kingdom*

### Effect of Thermodiffusive Instability on the Flame Topology of Premixed Ammonia/Hydrogen-Air Flames Anchored by a Bluff Body

Ahmed M. Elbanna[1], Tong Su[2], Yong Ren Tan[2], Epaminondas Mastorakos[2,3], Christine Rousselle[4,5]

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### Catalysis-Assisted Combustion for Improved Ammonia Flame Stability and Emission Control

Xuren Zhu[1], Yankun Du[3], Hao Hu[2], Peng Ma[1], Samir B. Rojas-Chávez[1], Jinzhou Li[1], Surendra Kumar Soni[1], Shumeng Xie[2], Qian He[3], Huangwei Zhang[1,2]

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### Scalar Characterization of Turbulent Premixed Partially Cracked Ammonia Swirling Flames

Samir B. Rojas-Chávez[1], Hao Hu[2], Peng Ma[1], Jinzhou Li[1], Surendra Kumar Soni[2], Shumeng Xie[2], Xuren Zhu[1], Huangwei Zhang[1,2]

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### Reaction Zone Stratification in Ammonia Swirl Flame with Pilot Injection of Partially Cracked Ammonia

Peng Ma[1], Hao Hu[2], Jinzhou Li[1], Xuren Zhu[1], Shumeng Xie[2], Surendra Kumar Soni[2], Samir B. Rojas-Chávez[1], Huangwei Zhang[1,2]

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### Spray-Flame Coupling and Reaction-Zone Structure in Swirl-Stabilized Ammonia-Methanol Combustion

Surendra Kumar Soni[1], Shumeng Xie[2], Samir B. Rojas-Chávez[1], Peng Ma[1], Jinzhou Li[1], Xuren Zhu[1], Hao Hu[2], Huangwei Zhang[2]

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### Ammonia Cracking Dynamics and Flame Response in an Integrated Catalytic Cracker and Swirl Burner System

Jinzhou Li[1], Limo He[1,3], Samir B. Rojas-Chávez[1], Peng Ma[1], Surendra Kumar Soni[2], Xuren Zhu[2], Shumeng Xie[2], Hao Hu[2], Ning Yan[1,3], Huangwei Zhang[1,2]

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### Numerical Study on Laminar Burning Velocity and Markstein Length of $NH_3/H_2$/Air Premixed Flames

Hao Shi[1], Akira Tsunoda[2], Xinyi Chen[3], Victor Lafaurie[1], Minh Bau Luong[4], Youhi Morii[2], Kaoru Maruta[2]

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[3] *School of Engineering, Newcastle University, United Kingdom*

[4] *School of Mechanical Engineering, Hanoi University of Science and Technology, Vietnam*

### Confinement Caused Error in Laminar Burning Velocity Measurement of Hydrogen-Air Mixtures

Shuya Iwasaki[1,2], Haru Okada[1,3], Youhi Morii[1,†], Kaoru Maruta[1,†]

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[†] *Visiting Professor, CARES, University of Cambridge, Singapore*

## Detonation, Ignition and Fire

### Peak Energy Wavenumber Effects on 3D Detonation–Turbulence Interaction

Hao Yan, Zehua Zhang, Xin Li, Shangpeng Li, Huangwei Zhang

*Department of Mechanical Engineering, National University of Singapore, Singapore*

### Effects of NO on One-Dimensional Pulsating Detonation

Xin Li, Huangwei Zhang

*Department of Mechanical Engineering, National University of Singapore, Singapore*

### Numerical Simulation of One-Dimensional Unsteady $H_2/O_2/Br_2$ Detonations with Detailed Chemistry

Zehua Zhang[1,2], Zifeng Weng[1], Huangwei Zhang[1,2]

[1] *Department of Mechanical Engineering, National University of Singapore, Singapore*

[2] *National University of Singapore (Chongqing) Research Institute, China*

### Non-Ideality and Radical Seeding in Ammonia/DME Droplet Autoignition

Shangpeng Li, Huangwei Zhang

*Department of Mechanical Engineering, National University of Singapore, Singapore*

### Flame Kernel Dynamics near the Lean Ignition Boundary in a Cavity-Stabilized Combustor

Yuxi Guo[1,2], Xiangchen Mao[2], Zhentan Yu[2], Fei Duan[1], Xiaomin He[2]

[1] *Energy Research Institute @ NTU, Nanyang Technological University, Singapore*

[2] *College of Energy and Power Engineering, Nanjing University of Aeronautics and Astronautics, China*

### Scale-Down Studies of Low Expansion Foam Qualification Tests Using Unsteady Pool Fires

Shivakumar Annaiappa, Sowriraajan Appathurai Venugopalan, Bhaskar Dixit
*Fire and Combustion Research Center, JAIN University, Bangalore, India*

## Plasma and Diagnostics

### Investigation of Plasma on Soot Formation in Ethylene Laminar Co-Flow Diffusion Flames Using Dielectric Barrier Discharge

Victor Lafaurie[1], Yong Ren Tan[1], Nader Shohdy[1], Hao Shi[1], Tomoya Suzuki[2], Markus Kraft[1], Youhi Morii[2], Kaoru Maruta[2]
[1] *Cambridge Centre for Advanced Research and Education in Singapore (CARES), Singapore*
[2] *Institute of Fluid Science, Tohoku University, Japan*

### Simultaneous Temperature and Electric Field Measurements via Rayleigh Scattering and E-FISH for Reduced Electric Field Quantification

Yang Zhijian, Edwin Setiadi Sugeng, Chng Tat Loon
*Department of Mechanical Engineering, College of Design and Engineering, National University of Singapore, Singapore*

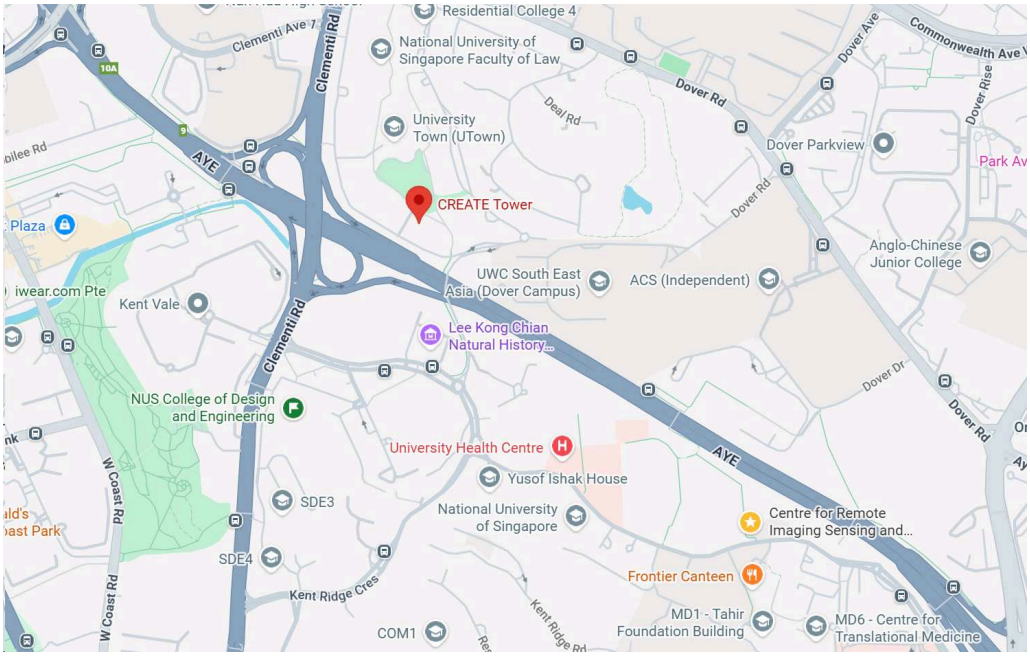
### Numerical Investigation of Non-Equilibrium Plasma Effects on FREI Dynamics in a Methane/Air Mixture within a Micro Flow Reactor with a Controlled Temperature

Tomoya Suzuki[1,2], Victor Lafaurie[1,3], Takuya Tezuka[1], Youhi Morii[1,†], Kaoru Maruta[1,†]
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# 5 Meeting Venue & Transportation Guide

**Venue:** Level 2 Theatrette, CREATE Tower, NUS University Town

**Address:** 1 CREATE Way, Singapore 138602




Map data © Google

## By Public Transportation

### Via MRT & NUS Internal Shuttle Bus

- **From Kent Ridge MRT Station (CC24):**
  Take Exit A to the bus stop at Kent Ridge Station Exit A / NUH (18331) and board the NUS Internal Shuttle Bus **D2**. Alight at the **UTown** stop, a short walk from CREATE Tower.

### Via MRT & Public Bus

- **From Buona Vista MRT Station (EW21/CC22):**
  Take Exit D to bus stop 11369 and board bus service **196**. Alight at bus stop 19059 (UTown, RC4), about 5 to 8 minutes' walk.
- **From Clementi MRT Station (EW23):**
  Take Exit B to bus stop 17179 and board bus service **183**, or board bus service **96** at Clementi Bus Interchange. Alight at bus stop 17099, about 8 to 10 minutes' walk.

# By Car or Taxi

**Driving Directions** Via the Ayer Rajah Expressway (AYE), take Exit 9 to Clementi Road, turn left into Kent Ridge Crescent (University Cultural Centre on your left), then follow College Link towards University Town. University Town can also be reached from Dover Road.

**Taxi / Private Hire Drop-off** Direct your driver to **"CREATE Tower, 1 CREATE Way"** (NUS University Town) and alight at the **CREATE Drop-off Point** along CREATE Way.

# Visitor Parking

- **Recommended Car Parks:** CREATE Tower car park, or the Stephen Riady Centre basement car park nearby if full.
- **White Lots Only:** Please park only in the **white lots**. Red lots are reserved for season pass holders.
- **Payment:** Parking fees are automatically charged via your in-vehicle unit (IU).

# 6 Acknowledgements

The Singapore Section of the Combustion Institute gratefully acknowledges the support of our sponsors.

## Industry Partners

## SME Partners

# About Our Industry Partners

## Analytical Technologies (LaVision)

Analytical Technologies is LaVision's local partner in Singapore, providing sales and technical support for its imaging systems. LaVision has supported the combustion community with advanced imaging solutions for more than 35 years. Its FlameMaster laser imaging systems deliver multi-parameter flame measurements with high spatial and temporal resolution, including volumetric 3D-LIF imaging of OH, while FlowBOS systems visualise reacting and non-reacting gas flows such as hydrogen and ammonia without seeding.

## Gaia Science

Gaia Science Pte Ltd is a Singapore-based scientific and engineering solutions provider serving the research, healthcare and industrial sectors across Southeast Asia. At this meeting, Gaia Science showcases Strobic high plume exhaust fans and industrial and laboratory scrubber systems for managing hazardous exhaust, emissions and combustion by-products safely and in compliance with regulations. Its specialists support customers from application assessment and system selection through installation, commissioning and preventive maintenance.

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Aegis Engineering & Consultancy Pte Ltd is a Singapore-based, bizSAFE Level 3 certified engineering firm. Positioning itself as "Your Preferred Business Partner", the company specialises in project management, hookup management, total facilities hookup and facility engineering, together with HSE and QA/QC consultancy and training. It also delivers high-purity and clean piping installation, with expertise in orbital GTAW, manual GTAW and SMAW welding.